



LUNDS
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Harmonic Analysis, Autumn 2026

MATP32

Tentative Course Schedule

- **Lecture 1 (01/09/2026).** In the first part of the lecture, a brief introduction to the contents of the course will be given. In the second part of the lecture, we will introduce maximal operators and study some of their basic properties.

Relevant material from [1]: §2.1.1.

- **Lecture 2 (04/09/2026).** In the first part of the lecture, we will prove the Marcinkiewicz interpolation theorem and then we will use it to establish L^p -boundedness of maximal operators for $1 < p \leq \infty$. In the second part of the lecture, we will discuss about some applications of boundedness of maximal operators to differentiation theory.

Relevant material from [1]: §1.3.1, §2.1.2, §2.1.3.

- **Lecture 3 (08/09/2026).** In this lecture, we will introduce approximate identities and briefly state some basic facts for Schwartz functions and tempered distributions that will be used in the course.

Relevant material from [1]: §2.2, §2.3.

- **Lectures 4-6 (11/09/2026, 15/09/2026, 18/09/2026).** We will introduce the Calderón-Zygmund decomposition for functions in L^1 . We will then establish the L^p -boundedness ($1 < p < \infty$) of a certain class of Calderón-Zygmund operators.

Relevant material from [1]: §5.3.1, §5.3.3.

- **Lectures 7-8 (22/09/2026, 25/09/2026).** We will introduce some classical Littlewood-Paley operators and establish their L^p -boundedness ($1 < p < \infty$).

Relevant material from [1]: §6.1.1, §6.1.3, and Appendix C.

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- **Lecture 9 (29/09/2026).** We will state various characterisations of the real Hardy space $H^1(\mathbb{R}^d)$ and we will prove that Calderón-Zygmund operators map $H^1(\mathbb{R}^d)$ to $L^1(\mathbb{R}^d)$ (using the characterisation of $H^1(\mathbb{R}^d)$ in terms of atoms).

Relevant material from [2] : §2.1.1, §2.4.1 (Theorem 2.4.1). We will also state results that correspond to Theorem 2.2.9 (for $p = 1$), Theorem 2.3.12 (for $p = 1$).

- **Lecture 10 (02/10/2020).** We will introduce the space $BMO(\mathbb{R}^d)$ and study some of its basic properties including the John-Nirenberg theorem. The duality between $H^1(\mathbb{R}^d)$ and $BMO(\mathbb{R}^d)$ will also be stated.

Relevant material from [2] : §3.1 and the statement of Theorem 3.2.2.

- **Lecture 11 (06/10/2026).** We will introduce the notion of Carleson measures and then study their connection with $BMO(\mathbb{R}^d)$.

Relevant material from [2] : §3.3.

- **Lecture 12 (09/10/2020).** We will introduce the system of Haar functions and prove a dyadic variant of the Littlewood-Paley theorem. We will also discuss about the characterisations of dyadic analogues of $H^1(\mathbb{R}^d)$ in terms of the integrability of dyadic square functions.

Relevant material from [1]: §6.4.1, §6.4.2, §6.4.3.

- **Lecture 13 (13/10/2020).** In this lecture, we will briefly introduce the notion of wavelets focusing on the case of the Haar wavelet.

Relevant material from [3]: §9.3–9.5.

- **Lectures 14, 15, 16 (16/10/2026, 20/10/2026, 23/10/2026) .** Project preparation

- **Lectures 17, 18 (times to be announced) .** Project presentations

Referenser

- [1] Grafakos, Loukas. *Classical Fourier analysis*. Third edition. Graduate Texts in Mathematics, 249. Springer, New York, 2014. xviii+638 pp.
- [2] Grafakos, Loukas. *Modern Fourier analysis*. Third edition. Graduate Texts in Mathematics, 250. Springer, New York, 2014. xviii+638 pp.
- [3] Pereyra, Maria Cristina; Ward, Lesley A. *Harmonic analysis. From Fourier to wavelets*. Student Mathematical Library, 63. IAS/Park City Mathematical Subseries. American Mathematical Society, Providence, RI; Institute for Advanced Study (IAS), Princeton, NJ, 2012. xxiv+410 pp.

Additional reading:

- [4] J. Duoandikoetxea, *Fourier analysis*, translated and revised from the 1995 Spanish original by David Cruz-Uribe, Graduate Studies in Mathematics, 29, Amer. Math. Soc., Providence, RI, 2001; MR1800316